

Problem Solving

1. Let $U \sim \text{Uniform}[5, 10]$, and let $Z = I_{\{U \in [5, 7)\}}$ and $Z_n = I_{\{U \in [5, 7+1/n^2)\}}$. Prove that $Z_n \rightarrow Z$ in probability.

2. Let $X_n \sim \text{Exponential}(n)$. Show that X_n converges to zero in probability.

3. Let $Z_n \sim \text{Uniform}[0, n]$, let $W_n = 5Z_n/(Z_n + 1)$, and let $W = 5$. Prove that W_n converges to W in probability.

4. Let Z_n be the sum of the squares of the numbers showing when we roll n fair dice. Find a number m such that $\frac{1}{n}Z_n$ converges to m in probability as $n \rightarrow \infty$.